

Swift Student Challenge 2026

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1. What problem is your app playground trying to solve and what inspired you to solve it?

I often travel with my family on long drives or ferry trips. In recent years, I noticed that my mother began experiencing motion sickness more frequently, and trips that once felt effortless started requiring more preparation.

It made me reflect on my own experience. I am also prone to motion sickness, but over time I developed a simple habit: I put on earphones, close my eyes, stay still, and listen to music. This routine has helped me avoid motion sickness for years.

Motion sickness is often described as a sensory conflict between visual input and the body's sense of balance. Reducing visual stimulation while maintaining a stable sense of orientation may ease that conflict.

Apple's technologies made this idea possible. Using motion data from iPhone through CoreMotion and spatial audio through earphones such as AirPods, I explored how sound could respond to movement.

The app calibrates a neutral position, places an anchor sound in space, and adjusts it with motion so the user perceives a stable horizon.

I named the project **Nagi**, a Japanese word describing the calm surface of the sea. The name reflects the experience I wanted to create: a small sense of stillness within movement.

2. Who would benefit from your app playground and how?

This app was created primarily for my mother.

Rather than relying solely on medication or extended rest, I wanted to design a lightweight tool they could try immediately during travel. The app guides users to close their eyes and maintain a stable head position while listening to audio designed with spatial characteristics.

Using spatial audio enabled earphones can create a subtle psychological sense of orientation and immersion. When the sound field feels stable and centered, it may reinforce a perception of balance. Combined with reduced visual input, this can help the user feel calmer during motion.

The app does not claim to treat or cure motion sickness. Instead, it offers a structured sensory experience that may help users feel more grounded during early discomfort.

While it began as something for my parents, it could potentially be useful for anyone who experiences motion-related discomfort during travel.

3. How did accessibility factor into your design process?

Because the app is intended for users who may already be experiencing discomfort, minimizing cognitive and visual load was a primary design goal.

The interface reduces required user actions to a minimum. Large controls, clear states, and simple navigation allow a session to begin quickly. Once activated, interaction is limited so that users can focus on resting rather than managing the app.

Visual elements are intentionally minimal. I carefully adjusted contrast and hierarchy to ensure that essential information stands out without introducing unnecessary visual stimulation. Buttons are large and clearly distinguishable, and the layout follows a straightforward structure to avoid confusion.

Color choices and spacing were selected to maintain a calm, uncluttered appearance. The overall goal was to remove friction (both visually and interactively) so that the experience feels intuitive and unobtrusive during moments of discomfort.

4. Did you use AI tools? Please specify.

Yes. I used generative AI tools selectively during development.

The core application logic, including motion data handling, session control, and audio interaction was designed and implemented by me in Swift. My primary focus was validating the underlying concept and building the technical structure independently.

AI tools supported complementary areas such as exploratory UI variations, temporary sound asset generation, and early-stage design iteration. In practice, this resembled collaborating with a multidisciplinary assistant: a front-end contributor, a sound ideation partner, and a documentation aide.

To work effectively with these tools, I had to define clear specifications, review outputs critically, and refine prompts iteratively. This process strengthened my ability to articulate requirements precisely and maintain structured documentation, similar to coordinating within a development team while retaining full ownership of the engineering decisions.

5. What other technologies did you use in your app playground, and why did you choose them?

The app was developed using SwiftUI to enable rapid iteration and a clear separation between UI and logic.

CoreMotion provides real-time device motion and orientation data, which is central to detecting movement patterns and informing the audio response.

AVFoundation manages audio playback and spatial characteristics, allowing responsive sound feedback tied to motion input.

Swift Concurrency was used to handle continuous sensor updates without blocking the interface, ensuring smooth performance during sessions.

These technologies were chosen because they allow sensor-driven interaction while maintaining a lightweight, responsive architecture appropriate for an experimental playground.

Beyond the Swift Student Challenge

At the Apple Developer Academy, I shared my project during an internal session with a small group of learners. It was the first app I had built independently, and I wanted to explain both the idea and what I learned during development.

I presented the key concept, the value the app aimed to deliver, and a Lo-Fi prototype to illustrate the experience. I then walked through the user flow and briefly explained the technical approach behind it.

The most meaningful part was the discussion that followed. Mentors and other learners shared feedback on clarity, interaction, and visual structure. Through that conversation, I learned how important it is to communicate intent clearly when presenting work.

Applying their suggestions helped me simplify parts of the interface and better align the design with my original idea. Later, when I showed the updated version of the app, the response was very encouraging.

That experience showed me how openly sharing progress, challenges, and ideas can strengthen both individual learning and the community around us.